AQGD gradient computations wrong for circuits using one parameter in >

https://github.com/qiskit-community/qiskit-aqua/issues/588

ROW 49

AQGD gradient computations wrong for circuits using one parameter in > 1 gate #588

Closed

xmax1 opened on Jul 3, 2019

Informations

- **Qiskit Aqua version**: 0.5.1
- **Python version**: 3.7.1
- **Operating system**: Ubuntu 18

What is the current behavior?

Computes wrong gradient for qaoa.
qiskit > aqua > components > optimizers > aqgd > deriv line 101
Computes the parameter shift rule for the params. Does not include the product rule for when one parameter is used for more than one gate.

Steps to reproduce the problem

Init qaoa circuit
Compute gradient
Compare with actual gradient (using, say, pennylane)

What is the expected behavior?

Uses product rule

Suggested solutions

Use the product rule

👍 1

chunfuchen added type: bug on Jul 3, 2019

chunfuchen on Jul 16, 2019 — Contributor

@Nick-Singstock do you have any thought?

woodsp-ibm assigned Cryoris on Jan 17, 2020

Cryoris added type: feature request on Mar 20, 2020

woodsp-ibm on Oct 22, 2020 — Member

@BryceFuller I believe the new gradient logic that has just been added has product rule support. Presumably this issue could be addressed by allowing AQGD to support a gradient function that was passed in rather than just ignore it as today. Any thoughts - would the gradient need to be typed checked here to ensure it was an analytic gradient? Is this something you might be able to take a look at?

pbark assigned adekusar-drl on Oct 27, 2020

pbark on Oct 27, 2020 — Collaborator

Taging also @adekusar-drl to give his opinion on this.

Assignees
Cryoris
adekusar-drl

Labels
type: bug
type: feature request

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

Notifications — Customize

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Participants
+1

None yet

Development

No branches or pull requests

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Participants

+1

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pbark on Oct 27, 2020 Collaborator

Taging also @adekusar-drl to give his opinion on this.

woodsp-ibm mentioned this on Jan 27, 2021

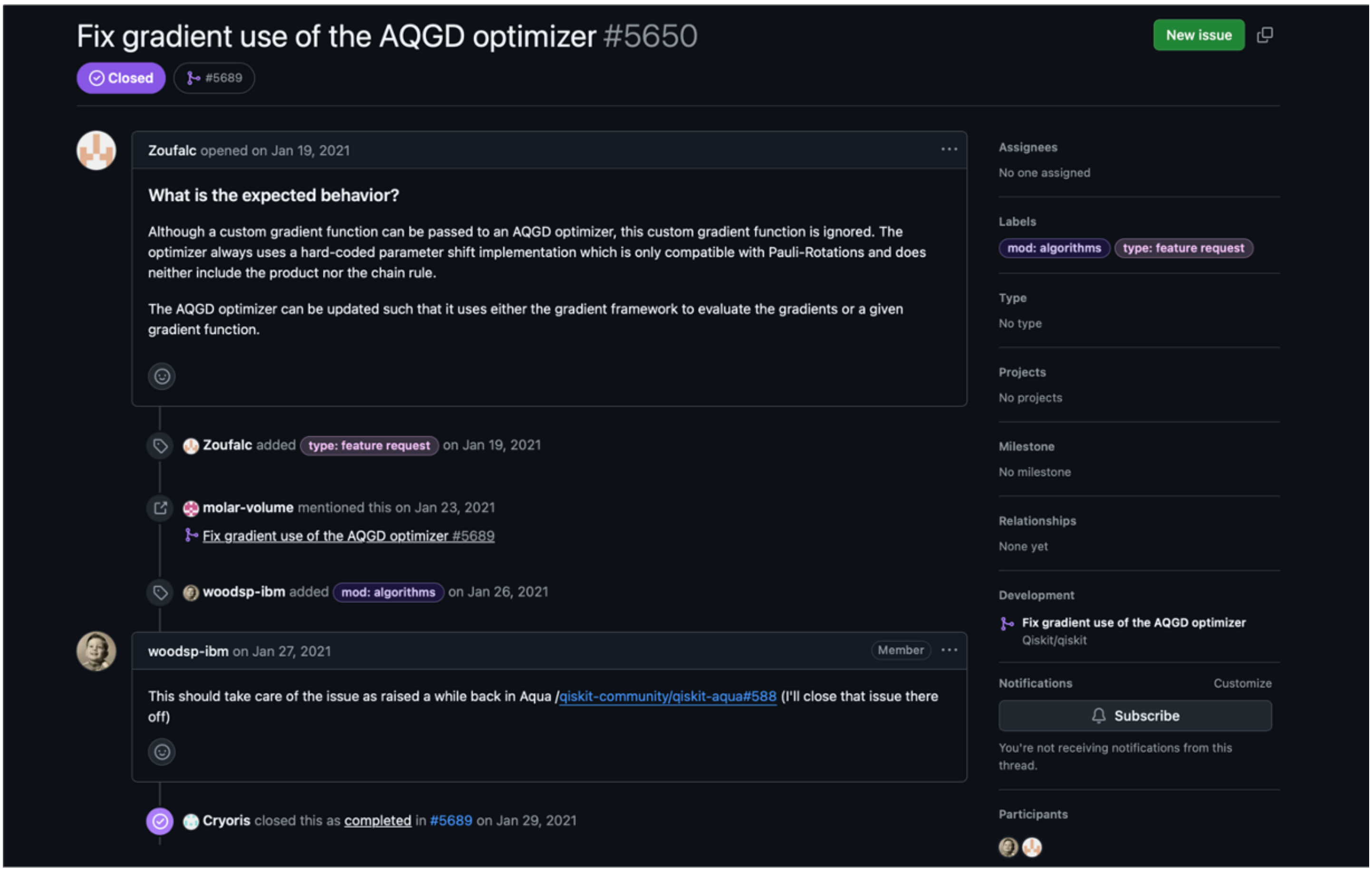
Fix gradient use of the AQGD optimizer Qiskit/qiskit#5650

woodsp-ibm on Jan 27, 2021 Member

Closing as there is a issue now opened in Terra to fix this in conjunction with the gradient framework that did not exist at the time.

👍 1

woodsp-ibm closed this as completed on Jan 27, 2021



Fix gradient use of the AQGD optimizer #5650

New issue

Closed · #5689

Zoufalc opened on Jan 19, 2021

What is the expected behavior?

Although a custom gradient function can be passed to an AQGD optimizer, this custom gradient function is ignored. The optimizer always uses a hard-coded parameter shift implementation which is only compatible with Pauli-Rotations and does neither include the product nor the chain rule.

The AQGD optimizer can be updated such that it uses either the gradient framework to evaluate the gradients or a given gradient function.

Zoufalc added type: feature request on Jan 19, 2021

molar-volume mentioned this on Jan 23, 2021

Fix gradient use of the AQGD optimizer #5689

woodsp-ibm added mod: algorithms on Jan 26, 2021

woodsp-ibm on Jan 27, 2021 · Member

This should take care of the issue as raised a while back in Aqua /qiskit-community/qiskit-aqua#588 (I'll close that issue there off)

Cryoris closed this as completed in #5689 on Jan 29, 2021

Assignees
No one assigned

Labels
mod: algorithms · type: feature request

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Fix gradient use of the AQGD optimizer
Qiskit/qiskit

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